

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18th June 2026
Student Name	Tirumala Srindhi
Project Name	ShopEZ -E-Commerce Application
Maximum Marks	4 Marks

Technical Architecture:

Shop-EZ uses a **three-tier architecture** — **Frontend, Backend, and Database** — ensuring scalability, security, and smooth data flow between components. Below is the detailed mapping of technologies used in each layer.

S.No	Component	Description	Technology
1	User Interface	Web UI for customers and admin dashboard for sellers	HTML, CSS, JavaScript, React.js
2	Application Logic-1	Handles user authentication, product management, and order processing	Node.js, Express.js
3	Application Logic-2	Manages secure payment and email confirmation services	Stripe API, Nodemailer
4	Application Logic-3	Provides analytics and reporting for sellers	Chart.js, Express middleware
5	Database	Stores collections for Users, Products, Cart, and Orders	MongoDB
6	Cloud Database	Cloud-hosted database for scalability and reliability	MongoDB Atlas
7	File Storage	Stores product images and banners	Cloudinary / Local filesystem
8	External API-1	Payment gateway integration	Stripe API
9	External API-2	Email confirmation and notifications	Gmail SMTP / Nodemailer
10	Machine Learning Model	Product recommendation engine (future enhancement)	TensorFlow.js
11	Infrastructure (Server / Cloud)	Deployment environment	AWS EC2 / Render Cloud / Localhost

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology	
1	Open-Source Frameworks	Uses open-source libraries for frontend and backend development	React.js, Node.js, Express.js	
2	Security Implementations	Implements encryption, authentication, and secure protocols	JWT, bcrypt, HTTPS, OWASP standards	
3	Scalable Architecture	Three-tier architecture supporting modular APIs and microservices	Node.js, MongoDB Atlas	
4	Availability	Ensures uptime through distributed servers and cloud hosting	AWS Load Balancer, MongoDB Atlas	
5	Performance	Optimized with caching and CDN for faster response	Redis Cache, Cloudflare CDN	